

Deploying Web App on ECS - Step 6: Verify ECS Task Definition

To Begin with the Lab

Summary of the Lab

In this step, the existing ECS task definition named my web app final was verified before creating the ECS service. The task definition was checked to make sure it used the correct Amazon ECR image URI for the PHP application, the proper task execution role, and the task role that allows the app to write to Amazon S3. The environment variables were also reviewed to confirm that the AWS Region and S3 bucket name matched the project setup. In addition, the CPU, memory, and port mapping were verified so the task could run correctly when the service is created. This confirmed that the task definition was ready for the next step, where the ECS Fargate service will be created.

Understanding ECS Task Definition Verification

An ECS task definition is the blueprint that tells Amazon ECS how to run a containerized application. It exists so ECS knows which image to pull, which IAM roles to use, which environment variables to pass, and how much CPU and memory to allocate. Verifying the task definition before creating a service is important because any mistake here can stop the app from starting correctly. For example, if the image URI or S3 bucket value is wrong, the application may launch but fail when it tries to upload files.

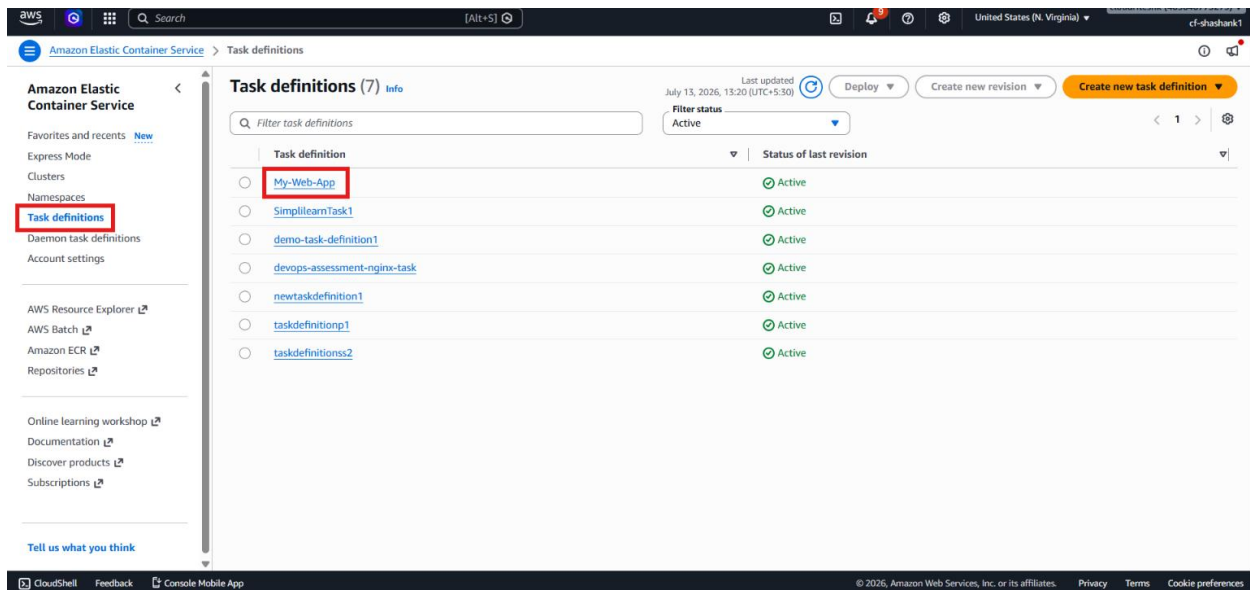
Example:

The task definition my web app final uses the ECR image URI for cloudfreeks-s3-php, the cloud app S3 access task role, and the AWS region plus S3 bucket environment variables. If the bucket name matches my web app new in ap-south-1, the app can upload files to the correct S3 bucket after it starts. This means the service can be created without fixing configuration errors later.

Lab Steps

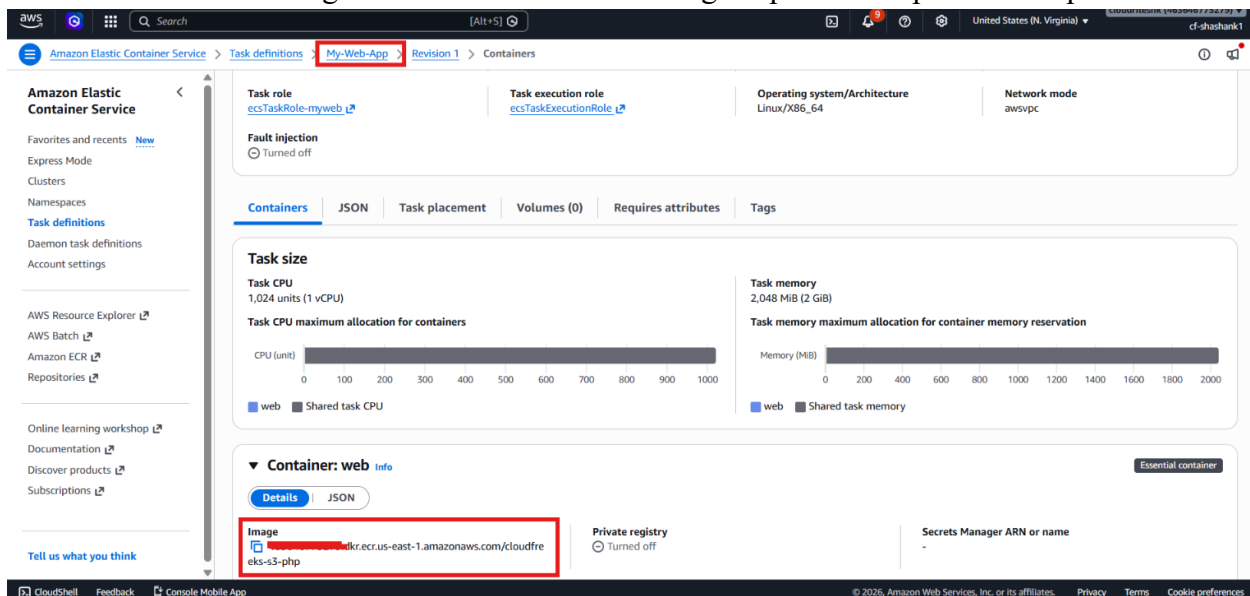
Step 1 – Open the ECS task definition

- Sign in to the **AWS Management Console**.
- Open **Amazon ECS**.
- Go to **Task definitions**.
- Open the task definition family named **My-Web-App**.



Step 2 – Verify the container image

- Check the container image field.
- Confirm that the image URI matches the ECR image copied in the previous step.



- Make sure the image points to the correct **cloudfreeks-s3-php** repository.

Step 3 – Verify the IAM roles

- Check the task role.

Amazon Elastic Container Service

My-Web-App:1

Overview

ARN: [arn:aws:ecs:us-east-1:123456789012:task-definition/My-Web-App:1](#)

Status: **ACTIVE**

Time created: July 9, 2026, 17:48 (UTC+5:30)

App environment: Fargate

Task role: [ecsTaskRole-myweb](#)

Task execution role: [ecsTaskExecutionRole](#)

Operating system/Architecture: Linux/X86_64

Network mode: awsvpc

Fault injection: Turned off

Containers

Task size

Task CPU: 1,024 units (1 vCPU)

Task CPU maximum allocation for containers

Task memory: 2,048 MiB (2 GiB)

Task memory maximum allocation for container memory reservation

- Confirm that the task role is access.
- Open the task role and verify that it has permission to write to Amazon S3.

Identity and Access Management (IAM)

ecsTaskRole-myweb

Allows ECS tasks to call AWS services on your behalf.

Summary

Creation date: July 09, 2026, 17:26 (UTC+05:30)

Last activity: 3 days ago

ARN: [arn:aws:iam::123456789012:role/ecsTaskRole-myweb](#)

Maximum session duration: 1 hour

Permissions

Permissions policies (1)

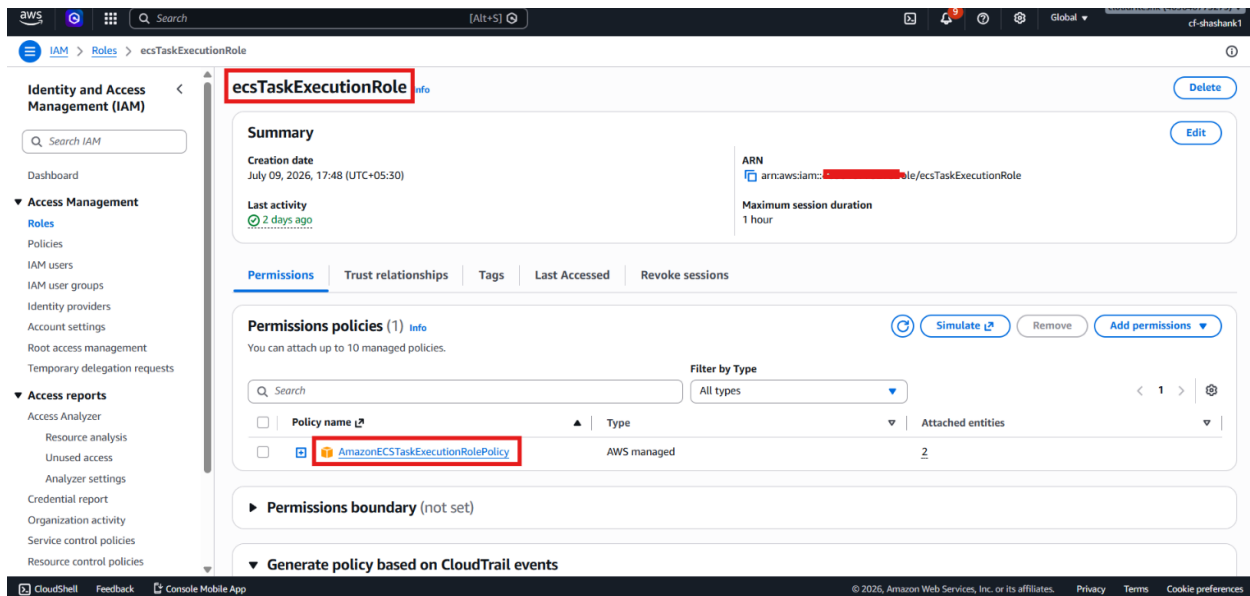
You can attach up to 10 managed policies.

Policy name: [AmazonS3FullAccess](#)

Type: AWS managed

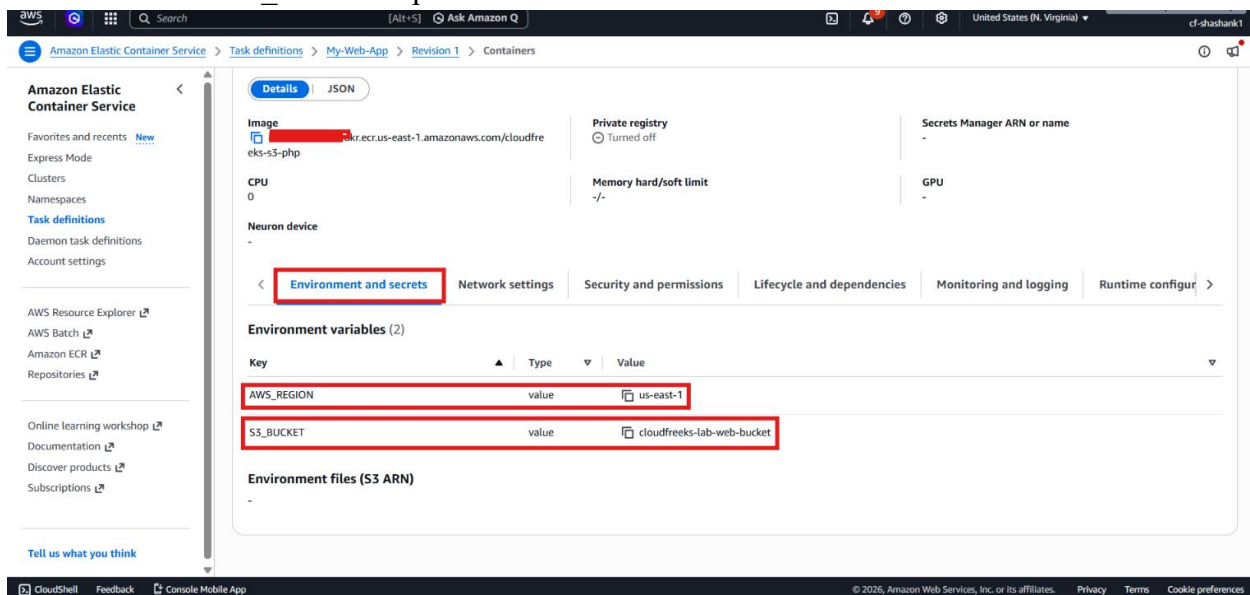
Attached entities: 42

- Check the task execution role.
- Confirm that the execution role is the ECS task execution role used to pull the image and send logs.



Step 4 – Verify the environment variables

- Open the environment variables section.
- Confirm that AWS_REGION is set to us-east-1.
- Confirm that S3_BUCKET points to the correct S3 bucket name.



- Check that the bucket exists in the same Region.

Step 5 – Verify task settings

- Check the CPU setting.
- Confirm that the task uses 1 vCPU.
- Check the memory setting.
- Confirm that the task uses 2 GB.

The screenshot shows the AWS Management Console for Amazon Elastic Container Service. The left sidebar contains navigation links for Favorites and recents, Express Mode, Clusters, Namespaces, Task definitions, Daemon task definitions, and Account settings. The main content area displays the 'My-Web-App:1' task definition. The 'Overview' tab is active, showing details like ARN, Status (ACTIVE), Time created, App environment (Fargate), Task role, Task execution role, Operating system/Architecture (Linux/x86_64), and Network mode (awsvpc). The 'Containers' tab is selected, showing the 'Task size' section. In this section, 'Task CPU' is set to 1,024 units (1 vCPU) and 'Task memory' is set to 2,048 MiB (2 GiB). Both values are highlighted with red boxes. Below these, there are progress bars for 'Task CPU maximum allocation for containers' and 'Task memory maximum allocation for container memory reservation'. The bottom of the console shows a footer with 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information.

- Check the port mapping.
- Confirm that the application runs on port 80.

The screenshot shows the AWS Management Console for Amazon Elastic Container Service, specifically the 'Container: web' details page. The left sidebar is the same as the previous screenshot. The main content area shows the 'Container: web' details. The 'Details' tab is active, showing fields for Image, CPU, Neuron device, Private registry, Memory hard/soft limit, Secrets Manager ARN or name, and GPU. The 'Network settings' tab is highlighted with a red box. Under 'Network settings', the 'Networking' status is 'Turned on'. Below this, the 'Port mappings (1)' section is shown. It contains a table with one row: 'Host port:Container port' 80:80. The 'Host port:Container port' text is highlighted with a red box. The table also shows the protocol as 'tcp', the port name as 'web-80-tcp', and the app protocol as 'http'. The bottom of the console shows a footer with 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information.

Step 6 – Confirm the task definition is ready

- Review all settings one last time.
- Confirm that the task definition is correctly configured for the web application.
- Prepare to use this task definition in the next step when creating the ECS service.

Verification

- Verified that the task definition family my web app final exists.
- Verified that the correct Amazon ECR image URI is attached.
- Verified that the task role has access to Amazon S3.

- Verified that the task execution role is configured for ECS image pulling and logging.
- Verified that `AWS_REGION` is set correctly.
- Verified that `S3_BUCKET` points to the correct S3 bucket.
- Verified that the CPU, memory, and port mapping are correct.
- Verified that the task definition is ready for ECS service creation.